

A trust-aware federated intrusion detection framework for privacy-preserving smart city IoT networks

Mehdi Houichi^{a,*}, Faouzi Jaidi^{a,b}, Adel Bouhoula^c

^a Innov'COM\Digital Security Research Lab, Higher School of Communications of Tunis (Sup'Com), University of Carthage, Tunis, Tunisia

^b National School of Engineers of Carthage, University of Carthage (ENICarthage), Tunis, Tunisia

^c Department of Next-Generation Computing, College of Graduate Studies, Arabian Gulf University, Manama, Kingdom of Bahrain

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ABSTRACT

The rapid deployment of Internet of Things (IoT) technologies in smart cities has significantly increased the attack surface of critical infrastructures, making efficient and privacy-preserving intrusion detection mechanisms essential. Federated learning (FL) has emerged as a promising paradigm for collaborative model training without sharing raw data; however, its performance degrades under non-independent and identically distributed (non-IID) data and adversarial settings, particularly in the presence of model poisoning attacks. In this paper, we propose a trust-aware federated intrusion detection framework that dynamically evaluates client reliability across communication rounds and integrates trust-based weighting into the aggregation process. The proposed approach combines validation-based trust scoring, outlier detection, and optional update clipping to mitigate malicious contributions while preserving model convergence under heterogeneous data distributions. The framework is evaluated on two benchmark intrusion detection datasets, CICIDS2017 and CSE-CIC-IDS2018, under non-IID conditions with varying levels of adversarial participation (20% and 40% malicious clients). Experimental results show that the proposed method provides improved robustness under high poisoning ratios, maintaining stable performance where conventional aggregation methods such as FedAvg degrade significantly. While statistical robust methods perform competitively under moderate attacks, the proposed trust-aware mechanism demonstrates superior resilience in more challenging adversarial scenarios. Furthermore, the proposed approach does not introduce additional communication overhead compared to standard federated learning, making it suitable for large-scale smart city deployments. These results highlight the effectiveness of trust-aware aggregation as a practical solution for secure and privacy-preserving collaborative intrusion detection.

1. Introduction

The rapid evolution of smart cities has led to the large-scale deployment of interconnected Internet of Things (IoT) devices and intelligent infrastructures, enabling advanced services across transportation, healthcare, energy, and public safety domains. While this pervasive connectivity enhances operational efficiency and data-driven decision-making, it simultaneously introduces significant cybersecurity risks due to the expanded attack surface and the heterogeneity of devices and communication protocols [1,2]. In this context, intrusion detection systems (IDS) play a critical role in identifying malicious activities and ensuring the resilience of smart city ecosystems. Traditional centralized IDS approaches rely on aggregating data from distributed sources into a

central server for model training and analysis [3]. However, such architectures face critical limitations, including high communication overhead, scalability challenges, increased latency, and, most importantly, privacy concerns associated with transmitting sensitive data across networks [4,5]. These limitations are particularly pronounced in smart city environments, where data originates from diverse and geographically distributed sources with strict privacy requirements. Federated learning (FL) has emerged as a promising paradigm to address these challenges by enabling collaborative model training without sharing raw data [6]. In FL, multiple clients train local models on their private datasets and share only model updates with a central aggregator, thereby preserving data privacy while benefiting from distributed learning [7]. Despite its advantages, FL introduces new challenges that

* Corresponding author.

E-mail addresses: mehdi.houichi@supcom.tn (M. Houichi), faouzi.jaidi@supcom.tn (F. Jaidi), a.bouhoula@agu.edu.bh (A. Bouhoula).

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are particularly relevant in smart city scenarios [8]. First, data generated by IoT devices is inherently non-independent and identically distributed (non-IID), reflecting variations in device types, user behavior, and network conditions. This heterogeneity can significantly degrade the convergence and generalization performance of federated models [9, 10]. Second, FL is vulnerable to adversarial attacks, including model poisoning and Byzantine behaviors, where malicious clients can manipulate local updates to disrupt the global model or degrade detection performance [11,12]. These challenges highlight the need for robust aggregation mechanisms capable of maintaining model integrity in adversarial and heterogeneous environments. To mitigate these issues, several robust aggregation methods have been proposed, including distance-based approaches such as Krum, coordinate-wise statistical methods such as Trimmed Mean and Median, and trust-based mechanisms that assign reliability scores to clients [13,14]. While these methods improve resilience to certain types of attacks, they often rely on static assumptions, lack adaptability over time, or fail to capture the dynamic reliability of clients in real-world deployments [15]. Moreover, many existing approaches are evaluated under simplified conditions and do not fully reflect the complexity of smart city environments characterized by non-IID data and varying levels of adversarial participation.

In this paper, we propose a trust-aware federated intrusion detection framework designed to enhance robustness and adaptability in smart city IoT networks. The proposed approach introduces a dynamic trust evaluation mechanism that continuously assesses client reliability across communication rounds based on validation consistency and deviation from expected update behavior. This trust score is integrated into the aggregation process to reduce the influence of potentially malicious or unreliable clients. Unlike conventional robust aggregation methods, the proposed framework combines multiple complementary strategies, including trust-based weighting, outlier detection, and optional update clipping, to provide enhanced resilience under diverse attack scenarios. The framework is specifically designed to operate under non-IID data distributions and varying adversarial conditions, making it suitable for realistic smart city deployments. The main contributions of this work can be summarized as follows:

- A trust-aware federated aggregation mechanism that dynamically evaluates client reliability over time, enabling adaptive mitigation of malicious contributions.
- A hybrid robust aggregation strategy combining trust weighting, outlier detection, and update clipping to improve resilience against poisoning attacks.
- A comprehensive experimental evaluation conducted on benchmark intrusion detection datasets (CICIDS2017 and CSE-CIC-IDS2018) under non-IID conditions with varying levels of adversarial participation.
- An analysis of robustness-performance trade-offs, demonstrating the effectiveness of the proposed approach particularly under high poisoning ratios while maintaining communication efficiency.

The remainder of this paper is organized as follows. [Section 2](#) reviews related work on federated learning and robust aggregation in intrusion detection. [Section 3](#) presents the system model and problem formulation. [Section 4](#) describes the proposed trust-aware aggregation framework. [Section 5](#) details the experimental setup and results. [Section 6](#) discusses the findings and limitations, and [Section 7](#) concludes the paper.

2. Related work

2.1. Centralized smart city IDS

Centralized intrusion detection systems remain widely used in smart city environments because they enable unified monitoring, centralized policy enforcement, and global visibility over distributed IoT traffic

[16]. In these architectures, traffic generated by heterogeneous urban subsystems, such as transportation networks, smart grids, industrial control systems, healthcare platforms, and public surveillance infrastructures, is collected and analyzed within a central Security Operations Center (SOC). Deep learning-based centralized IDS models, including CNN-, LSTM-, and hybrid-based approaches, have reported strong detection performance in controlled settings [17]. More recent studies have further enhanced centralized IDS through transformer-based architectures and graph neural networks for large-scale IoT traffic analysis [18]. Despite these advances, centralized IDS architectures face structural limitations in realistic smart city deployments. Privacy regulations and data protection requirements may restrict the large-scale centralization of traffic traces [19]. Continuous forwarding of raw traffic from geographically distributed gateways also introduces substantial bandwidth, storage, and latency costs [20]. In addition, the central server constitutes both a high-value target and a single point of failure. These limitations have motivated a growing shift toward distributed and edge-assisted security analytics in urban environments [21].

2.2. Federated learning for IoT and smart city security

Federated learning, introduced by McMahan et al., enables collaborative model training across distributed clients while preserving data locality, typically through Federated Averaging (FedAvg) [22]. This paradigm is especially attractive for smart city cybersecurity because it reduces direct privacy exposure by keeping raw traffic data at the edge while still allowing joint model optimization across multiple urban districts or services. Recent studies have explored FL-based intrusion detection in distributed IoT and smart city settings, showing improvements in scalability and privacy preservation compared with centralized baselines [23]. However, practical deployment of FL in smart city environments is complicated by two major forms of heterogeneity. First, statistical heterogeneity arises because traffic distributions vary significantly across districts, services, device types, and local attack patterns. Recent work on personalized and clustered FL attempts to mitigate this issue, but these approaches may weaken global generalization, which remains important for detecting coordinated or cross-domain attacks [24]. Second, system heterogeneity results from variations in computation capacity, participation frequency, and network quality across edge nodes. Although asynchronous FL and client-selection strategies can reduce efficiency bottlenecks, they do not directly solve the robustness problem under malicious participation [25]. As a result, while FL addresses privacy and scalability concerns, it still faces major challenges in convergence stability and adversarial resilience in heterogeneous smart city IoT environments [26].

2.3. Byzantine-resilient and robust aggregation mechanisms

To improve federated learning robustness under adversarial participation, a large body of work has focused on robust aggregation rules. Byzantine-resilient methods such as Krum and Multi-Krum select updates that are closest to the majority in parameter space, thereby filtering extreme outliers under bounded adversarial assumptions [27]. Coordinate-wise aggregation rules such as Median and Trimmed Mean reduce the effect of abnormal parameter values dimension by dimension and remain popular due to their simplicity and theoretical tractability [28]. More recent studies have expanded this family through gradient clipping, anomaly-aware aggregation, similarity filtering, and hybrid robust defenses [29].

Nevertheless, these methods share an important limitation: they often assume that benign client updates form a reasonably coherent cluster and that malicious updates appear as detectable outliers [30]. In smart city intrusion detection, this assumption may not hold because benign clients naturally experience different traffic profiles, dominant attack families, and service-specific distributions. Under such non-IID

conditions, legitimate updates may differ substantially in direction and magnitude, which can cause statistical robust rules to over-filter benign clients or fail to capture stealthy poisoning behavior [31,32]. Therefore, although robust aggregation improves resilience in many settings, purely statistical filtering may be insufficient in highly heterogeneous smart city environments.

2.4. Trust-based federated learning and reputation modeling

Trust-aware federated learning introduces a more adaptive perspective by estimating client reliability over time instead of relying only on geometric or coordinate-wise deviation metrics [33]. Representative approaches include FoolsGold, which mitigates sybil-style attacks by penalizing highly similar gradient directions [34], and FLTrust, which derives trust scores from alignment with a trusted server-side reference dataset [35]. More recent work has extended trust modeling through temporal consistency analysis, contribution-aware weighting, stability-based reputation, and blockchain-supported trust recording [36].

Despite their promise, current trust-based FL approaches still exhibit several limitations. Some methods depend on a trusted server-side dataset, which may be impractical in privacy-sensitive smart city settings [37]. Others are designed for specific attacker models, such as sybil collusion or extreme-gradient manipulation, which limits generalization against adaptive poisoning strategies. In addition, relatively few studies evaluate trust-aware FL under realistic smart city non-IID intrusion detection scenarios, where benign divergence is frequent and may be mistaken for malicious behavior [38]. Domain-specific operational constraints, such as maintaining stable detection thresholds and avoiding excessive false positives, are also not sufficiently addressed in many existing trust formulations [39]. These limitations indicate that trust mechanisms for smart city IDS should be more context-aware, lightweight, and robust to both heterogeneity and evolving adversarial behavior.

2.5. Secure aggregation and privacy enhancement extensions

Beyond robustness, federated learning research has increasingly incorporated privacy-enhancing mechanisms such as secure aggregation, differential privacy, and homomorphic encryption [40–41]. These techniques strengthen confidentiality by preventing direct exposure of individual client updates and reducing risks such as membership inference or parameter reconstruction [42]. Such extensions are particularly relevant in smart city deployments, where traffic data may contain sensitive citizen, infrastructure, or operational information.

However, these techniques introduce their own trade-offs. Additional encryption, noise injection, or secure computation layers may increase communication payload, latency, and computational cost, which can be problematic for resource-constrained IoT gateways and real-time security applications [43]. More importantly, privacy enhancement alone does not ensure robustness: encrypted or privatized malicious updates may still corrupt the global model if no adversarial filtering mechanism is applied [44]. For this reason, recent literature increasingly emphasizes that confidentiality and robustness should be addressed jointly rather than separately [45]. In addition to software-level robustness mechanisms, recent research has highlighted the importance of lightweight cryptographic protection, hardware-assisted trust, and secure acceleration architectures for protecting distributed learning and IoT infrastructures in smart city environments. Efficient implementations of elliptic-curve cryptography, secure digital signatures, and post-quantum cryptographic accelerators have been proposed to improve authentication, integrity verification, and secure communication for resource-constrained edge devices and embedded systems. Recent studies on lightweight Ed25519 architectures, FPGA-based cryptographic accelerators, SIKE implementations, and polynomial multiplication accelerators for post-quantum

cryptography demonstrate the growing importance of integrating efficient security primitives into next-generation distributed systems [46–47]. Similarly, fault-detection and hardware-level protection mechanisms have been investigated to improve resilience against side-channel attacks and implementation-level vulnerabilities in secure IoT infrastructures [48,49]. Although the present work focuses primarily on robust trust-aware aggregation against poisoning attacks, these cryptographic and hardware-assisted security mechanisms represent important complementary directions for future secure federated intrusion detection deployments in smart city environments.

2.6. Gap summary and positioning

The existing literature provides important but fragmented building blocks for secure federated intrusion detection. Centralized IDS offers strong global visibility but suffers from privacy, scalability, and latency limitations [16–21]. Federated learning addresses raw-data centralization but remains vulnerable to heterogeneity and poisoning attacks [22–26]. Robust aggregation methods improve resilience under bounded adversarial assumptions, yet they often struggle when benign non-IID divergence resembles malicious behavior [27–32]. Trust-based methods introduce adaptive reliability estimation, but many still depend on restrictive assumptions, narrow attack models, or evaluation settings that do not fully capture the operational complexity of smart city intrusion detection [33–39]. Privacy-enhancing techniques further strengthen confidentiality, but they do not directly solve adversarial robustness [40–45]. Accordingly, an important research gap remains in designing a federated intrusion detection framework that simultaneously addresses four requirements: privacy preservation, robustness to malicious or unreliable participants, stability under non-IID smart city traffic, and operational feasibility for large-scale urban deployments. This gap motivates the framework proposed in this paper, which combines dynamic trust-aware client weighting with robust aggregation principles to improve resilience and convergence stability in heterogeneous and potentially compromised smart city IoT networks.

3. System model and threat model

3.1. Smart city architecture

We consider a smart city environment composed of geographically distributed urban districts, each hosting heterogeneous IoT subsystems such as intelligent transportation sensing, video surveillance, smart metering, healthcare monitoring, and environmental data collection. To secure this ecosystem, the proposed architecture adopts a four-layer hierarchical design, as illustrated in Fig. 1. At the IoT Device Layer, a large number of resource-constrained devices continuously generate diverse network flows—including telemetry data, control signals, and multimedia streams—over wired and wireless infrastructures; these devices are assumed to be potentially vulnerable to compromise and may participate in coordinated attack campaigns such as botnets or distributed denial-of-service attacks. At the Edge Gateway Layer, each district deploys a gateway that aggregates local traffic, performs feature extraction, and trains a district-specific intrusion detection model using only locally available data; these gateways operate as federated learning clients and never transmit raw traffic outside their administrative boundaries, thereby preserving privacy and reducing bandwidth usage. Above this layer, the Federated Server Layer coordinates the collaborative learning process by orchestrating training rounds, selecting participating edge clients, collecting model updates, and performing trust-aware aggregation to generate a robust global intrusion detection model; it also integrates a trust management module that dynamically evaluates client reliability across rounds. Finally, at the Security Operations Center (SOC) layer, the global model is deployed for centralized monitoring, alert management, triage, and mitigation actions; the SOC receives aggregated security intelligence rather than raw traffic data and

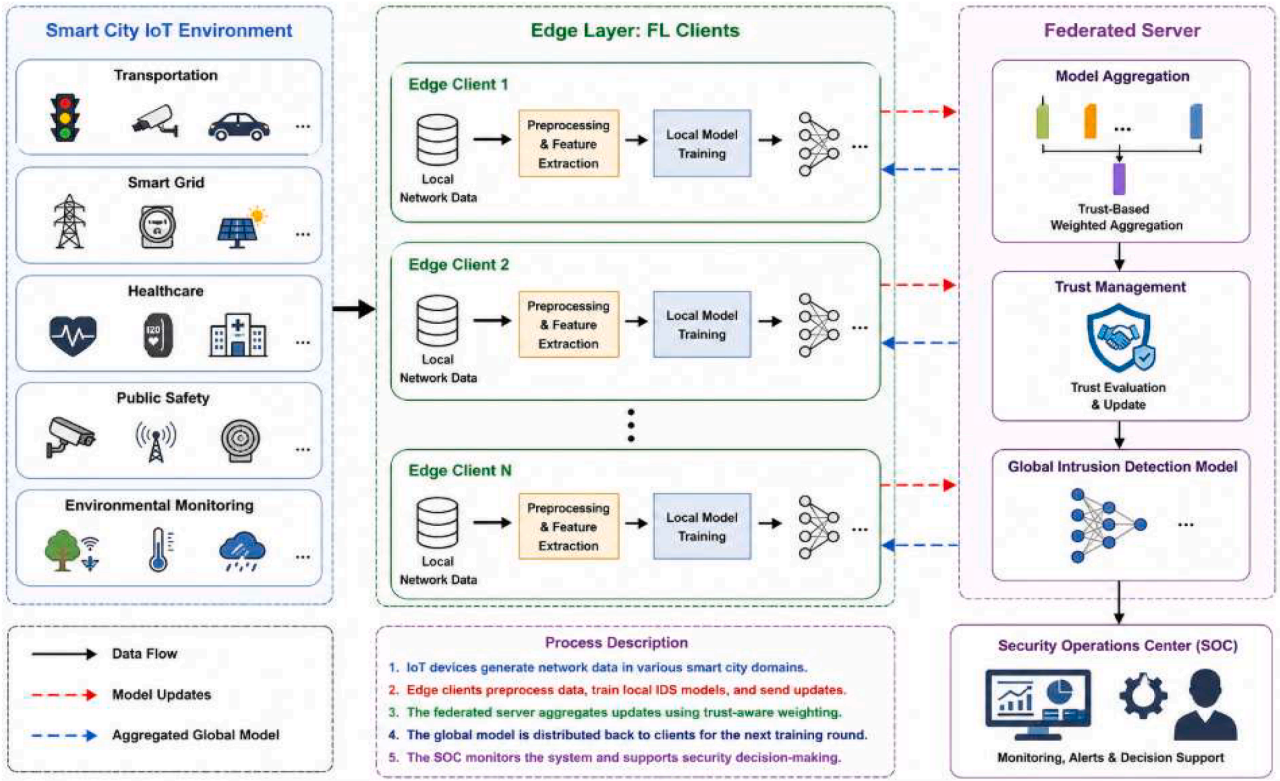


Fig. 1. Four-Layer Hierarchical Architecture of the Proposed Trust-Weighted Federated IDS Framework.

supervises the training lifecycle through high-level performance tracking and policy enforcement. Fig. 1 depicts the end-to-end communication flow across these layers, highlighting how IoT devices generate traffic, edge gateways conduct local training, the federated server performs trust-weighted aggregation, and the SOC oversees the global security posture of the smart city.

3.2. System model

We consider a federated learning framework for intrusion detection in smart city IoT environments, where multiple distributed clients collaboratively train a shared global model without exchanging raw data. These clients may represent edge gateways, network domains, or IoT clusters deployed across different urban services. Let K denote the total number of participating clients, where each client $k \in \{1, \dots, K\}$ holds a local dataset D_k of size n_k . These datasets consist of labeled network traffic records used to train intrusion detection models. Due to the inherent diversity of smart city infrastructures, the local datasets are generally heterogeneous and follow non-independent and identically distributed (non-IID) patterns. The objective of federated learning is to jointly optimize a global model parameterized by θ , by minimizing the weighted empirical risk across all clients:

$$F(\theta) = \sum_{k=1}^K \frac{n_k}{n} F_k(\theta) \quad (1)$$

where $n = \sum_{k=1}^K n_k$ represents the total number of training samples, and $F(\theta)$ denotes the local loss function at client k .

Training proceeds in communication rounds. At each round t , the central server selects a subset of clients $S^t \subseteq \{1, \dots, K\}$ to participate. Each selected client performs local training using its private dataset and computes a model update Δ_k^t , defined as the difference between the locally updated model θ_k^t and the current global model θ^t :

$$\Delta_k^t = \theta_k^t - \theta^t \quad (2)$$

In conventional federated learning, the global model is updated using weighted averaging (FedAvg), where each client contribution is proportional to its dataset size:

$$\theta^{t+1} = \theta^t + \sum_{k \in S^t} \frac{n_k}{\sum_{j \in S^t} n_j} \Delta_k^t \quad (3)$$

While this approach is effective under benign conditions, it assumes that all participating clients provide reliable updates, which is not guaranteed in adversarial smart city environments.

3.3. Problem statement

In real-world smart city IoT deployments, federated learning is challenged by both statistical heterogeneity and adversarial behavior. First, the distribution of local datasets varies significantly across clients due to differences in network traffic patterns, device characteristics, and localized attack scenarios. This non-IID nature of data can slow down convergence and degrade the generalization capability of the global model. Second, federated learning is vulnerable to adversarial manipulation. A subset of participating clients may act maliciously and submit manipulated updates with the objective of degrading model performance or introducing hidden biases. Let $M^t \subseteq S^t$ denote the set of malicious clients at communication round t , such that:

$$|M^t| = \alpha |S^t| \quad (4)$$

where α represents the proportion of malicious participants. Under these conditions, standard aggregation methods such as FedAvg become unreliable, as they treat all client updates equally and cannot distinguish between benign and malicious contributions.

3.4. Threat model

We adopt a Byzantine threat model in which malicious clients can arbitrarily manipulate their updates before sending them to the central server. These adversarial clients may perform various types of attacks, including label flipping, gradient scaling, random perturbation, or more sophisticated model poisoning strategies. The adversary is assumed to have the ability to modify local training behavior and craft updates that aim to either maximize degradation of the global model performance or remain stealthy to avoid detection. Malicious clients may persist across multiple communication rounds and may coordinate their behavior to increase the effectiveness of the attack. We assume that the central server does not have prior knowledge of which clients are malicious and does not rely on any trusted reference dataset. Honest clients follow the prescribed training protocol and generate updates based solely on their local data. The proportion of malicious clients may vary across scenarios, including moderate and high adversarial settings.

3.5. Design objective

Given the challenges of non-IID data and adversarial participation, the objective of this work is to design a robust aggregation mechanism

that can effectively mitigate the influence of malicious updates while preserving learning performance. Specifically, the proposed approach aims to dynamically evaluate the reliability of participating clients, reduce the contribution of suspicious updates, and maintain stable convergence under heterogeneous data distributions. At the same time, the solution must remain communication-efficient and scalable, making it suitable for large-scale smart city deployments. To address these requirements, we introduce a trust-aware aggregation mechanism that integrates dynamic trust evaluation into the federated learning process, enabling adaptive and robust model aggregation in adversarial environments.

4. Proposed trust-aware aggregation framework

4.1. Overview of the proposed framework

To address the challenges of non-IID data and adversarial participation in federated intrusion detection, we propose a trust-aware aggregation framework that dynamically evaluates client reliability and integrates trust scores into the aggregation process.

The overall system architecture is illustrated in Fig. 1, where multiple distributed clients (e.g., IoT gateways or edge nodes) perform local

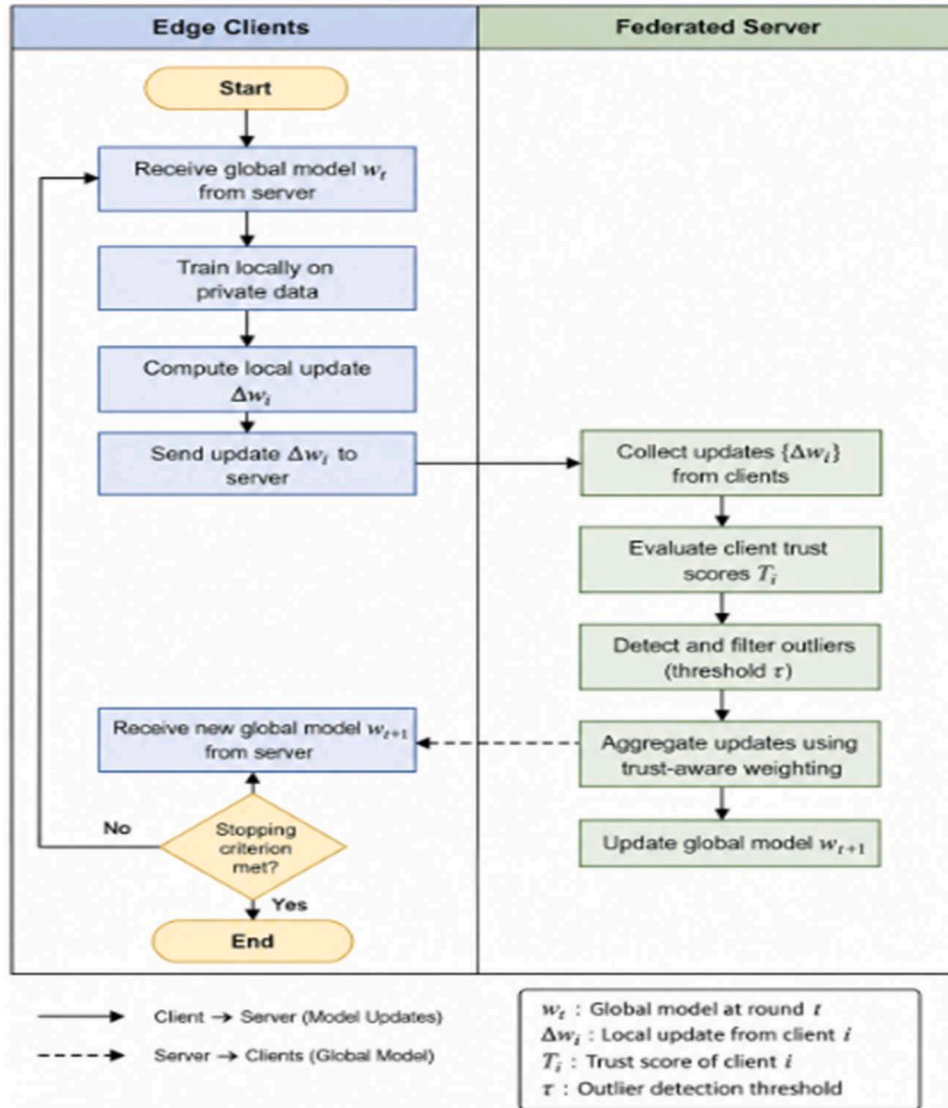


Fig. 2. Trust-Aware Federated Intrusion Detection Architecture for Smart Cities.

model training on private datasets and send their updates to a central server. At each communication round, the server collects local updates from participating clients and applies a multi-step aggregation strategy consisting of trust evaluation, outlier detection, and trust-weighted aggregation. The detailed workflow of this process is depicted in Fig. 2, which highlights how client updates are processed and filtered before contributing to the global model. This layered design allows the framework to mitigate malicious behavior while preserving useful information from heterogeneous clients.

4.2. Trust score computation

The core of the proposed method is a dynamic trust evaluation mechanism that assigns a reliability score to each participating client. Let $\tau_k^t \in [0,1]$ denote the trust score of client k at communication round t . This score evolves over time to reflect both historical behavior and current update quality. The trust score is updated using an exponential smoothing formulation:

$$\tau_k^t = \lambda \tau_k^{t-1} + (1 - \lambda) s_k^t \quad (5)$$

where $\lambda \in [0,1]$ is a memory factor controlling the influence of past trust values, and s_k^t is the instantaneous trust score computed at round t . The instantaneous trust score combines two complementary components: validation consistency and update deviation. It is defined as:

$$s_k^t = \alpha v_k^t + (1 - \alpha) d_k^t \quad (6)$$

where $\alpha \in [0,1]$ balances the contribution of each component.

The validation consistency term v_k^t reflects the effectiveness of a client's update in improving model performance on a validation set or proxy evaluation mechanism. It is normalized relative to other participating clients to ensure comparability. The deviation term d_k^t measures how close a client's update is to the majority behavior, typically computed as the distance between the client update and a reference update such as the median. An exponential transformation is applied to penalize large deviations while preserving sensitivity to moderate differences. This combination enables the framework to distinguish between benign heterogeneity and potentially malicious behavior.

4.3. Outlier detection and update filtering

In addition to trust scoring, the framework incorporates an outlier detection mechanism to identify suspicious client updates. At each round, a reference update is computed (e.g., coordinate-wise median), and the distance between each client update and this reference is evaluated. A client update is considered an outlier if its deviation exceeds a predefined threshold. Such updates can either be discarded or assigned a significantly reduced trust score. This step is particularly important in scenarios involving strong adversarial behavior, where malicious updates may attempt to deviate significantly from the benign majority. To further enhance robustness, an optional clipping mechanism is applied to limit the magnitude of client updates. This prevents adversarial clients from dominating the aggregation process by submitting excessively large gradients. Together, outlier detection and clipping act as complementary safeguards that reinforce the trust-based evaluation.

4.4. Trust-aware aggregation rule

After computing trust scores and filtering suspicious updates, the global model is updated using a trust-weighted aggregation rule. Unlike standard federated averaging, where all clients are treated equally based on dataset size, the proposed method assigns higher importance to more reliable clients. The global model update is defined as:

$$\theta^{t+1} = \theta^t + \sum_{k \in S^t} w_k^t \Delta_k^t \quad (7)$$

where the aggregation weights w_k^t are computed as:

$$w_k^t = \frac{\tau_k^t \cdot n_k}{\sum_{j \in S^t} \tau_j^t n_j} \quad (8)$$

This formulation ensures that both data contribution and client reliability are taken into account during aggregation. As a result, updates from highly trusted clients have a stronger influence on the global model, while the impact of unreliable or malicious clients is progressively reduced.

4.5. Algorithm: trust-aware federated aggregation

The overall procedure of the proposed trust-aware federated aggregation can be summarized as follows. At each communication round, a subset of clients is selected to participate in training. Each client performs local model updates using its private dataset and sends the resulting update to the central server. The server then computes a reference update and evaluates each client based on validation consistency and deviation from the reference. Trust scores are updated dynamically, and outlier updates are filtered or down-weighted. Finally, the global model is updated using trust-weighted aggregation, optionally combined with update clipping to ensure stability. This integrated approach enables the framework to adapt to changing client behavior over time while maintaining robustness against adversarial manipulation and preserving learning performance under heterogeneous data distributions (Algorithm 1).

5. Experimental results and analysis

The experimental evaluation is conducted using two widely adopted benchmark datasets for network intrusion detection: CICIDS2017 and CSE-CIC-IDS2018. These datasets provide realistic network traffic scenarios, including both benign and malicious activities, and have been extensively used to evaluate machine learning-based intrusion detection systems. The CICIDS2017 dataset contains diverse attack categories such as DoS, DDoS, brute-force, and infiltration attacks, along with normal traffic, offering a balanced representation of common cybersecurity threats. In contrast, CSE-CIC-IDS2018 presents a more complex and large-scale environment with higher traffic variability, increased class imbalance, and more sophisticated attack patterns, making it particularly suitable for evaluating robustness under heterogeneous conditions. Prior to training, standard preprocessing steps are applied, including feature normalization and removal of non-informative attributes. The datasets are partitioned across multiple clients in a non-independent and identically distributed (non-IID) manner to simulate realistic smart city environments, where each client observes different traffic distributions and attack types.

5.1. Results on CICIDS2017

5.1.1. Convergence behavior and stability

The convergence dynamics on the CICIDS2017 dataset are illustrated in Figs. 3 and 4. Under clean conditions, FedAvg achieves rapid convergence and reaches high accuracy ($\approx 98.4\%$), confirming its efficiency in homogeneous and non-adversarial settings. However, when 20% of clients are malicious, FedAvg exhibits clear oscillatory behavior across rounds, reflecting its sensitivity to poisoned gradients and its inability to distinguish between benign and adversarial updates. Robust statistical methods such as Trimmed Mean and coordinate-wise Median mitigate extreme deviations and reduce instability compared to poisoned FedAvg. As shown in Fig. 3, these methods maintain relatively stable trajectories, yet moderate fluctuations persist due to the interaction between non-IID data distributions and partially stealthy adversarial updates that do not appear as extreme outliers.

In contrast, the proposed trust-aware aggregation exhibits a distinct

Algorithm 1

Trust-Aware Federated Aggregation.

Input:

- Initial model θ^0
- Trust scores $\tau_k^0 = 1$
- Memory factor λ
- Threshold τ
- Clipping bound C

- 1: For each round $t = 1$ to T :
- 2: Select client subset S^t
- 3: For each client $k \in S^t$:
- 4: Train local model $\rightarrow \theta_k^t$
- 5: Compute update Δ_k^t
- 6: Compute reference update (median)
- 7: For each client $k \in S^t$:
- 8: Compute validation score v_k^t
- 9: Compute deviation score d_k^t
- 10: Compute $s_k^t = \alpha \cdot v_k^t + (1-\alpha) \cdot d_k^t$
- 11: Update trust τ_k^t
- 12: If update is outlier:
- 13: reduce trust or discard
- 14: Apply clipping to Δ_k^t
- 15: Compute weights w_{k^t} using trust
- 16: Aggregate:
- 17: $\theta^{t+1} = \theta^t + \sum_{k \in S^t} w_k^t \Delta_k^t$
- 18: Return final model θ^t

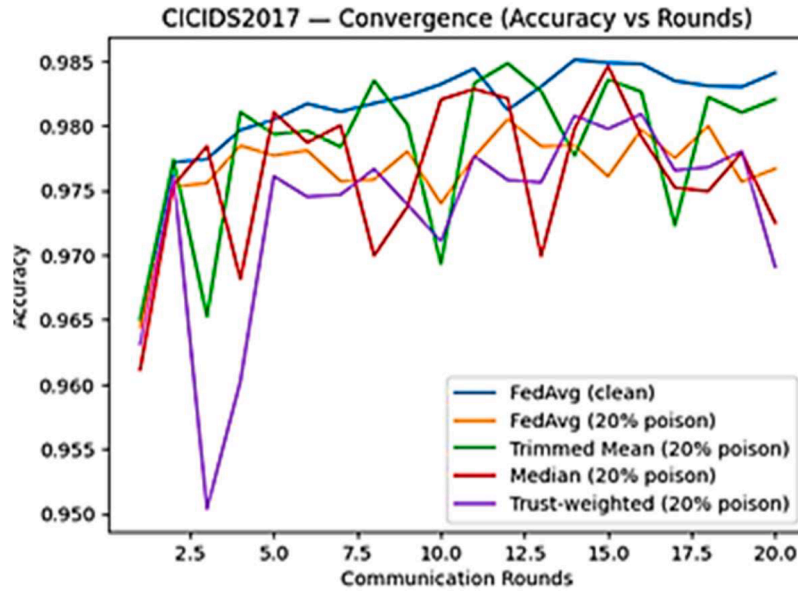


Fig. 3. Accuracy vs. Communication Rounds.

convergence pattern. As observed in Figs. 3 and 4, the method introduces a short calibration phase during early rounds, where performance increases more gradually compared to baseline methods.

This behavior is directly linked to the dynamic estimation of client trust scores, which initially reduces effective aggregation weights while the system learns reliability patterns. After this phase, the model stabilizes and maintains consistent performance with reduced variance. This result highlights a fundamental trade-off: while statistical methods optimize short-term convergence, trust-aware aggregation prioritizes long-term stability by progressively suppressing unreliable contributions.

5.1.2. Trust dynamics and client reliability

The evolution of client trust scores is presented in Fig. 5, which provides key insight into the internal behavior of the proposed framework. The results show a clear divergence between benign and malicious

clients over communication rounds. Malicious clients experience a rapid and sustained decrease in trust, approaching near-zero values, while benign clients maintain or recover high trust levels.

This behavior demonstrates that the proposed trust mechanism effectively performs implicit client reliability estimation without requiring external supervision or trusted validation data. Unlike hard filtering approaches, the trust-aware method does not abruptly discard clients; instead, it gradually reduces their influence, thereby preserving training stability. The smoothing mechanism further prevents abrupt oscillations, ensuring that temporary deviations due to non-IID heterogeneity do not lead to premature penalization of benign clients.

5.1.3. Robustness under increasing adversarial participation

The robustness of different aggregation strategies is summarized in Fig. 6 and Table 2. Under clean conditions (0% malicious clients), all methods achieve comparable performance (~ 0.95 F1-score), indicating

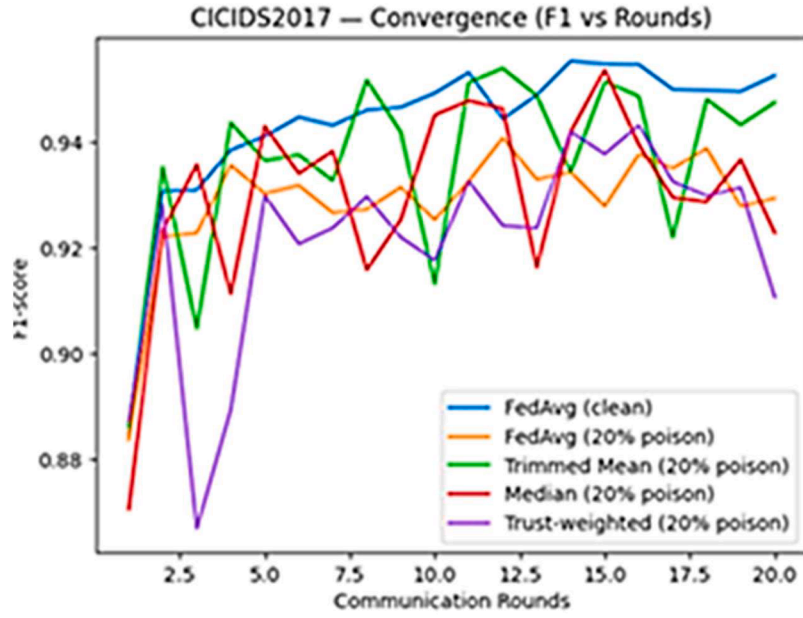


Fig. 4. F1-score vs. Communication Rounds.

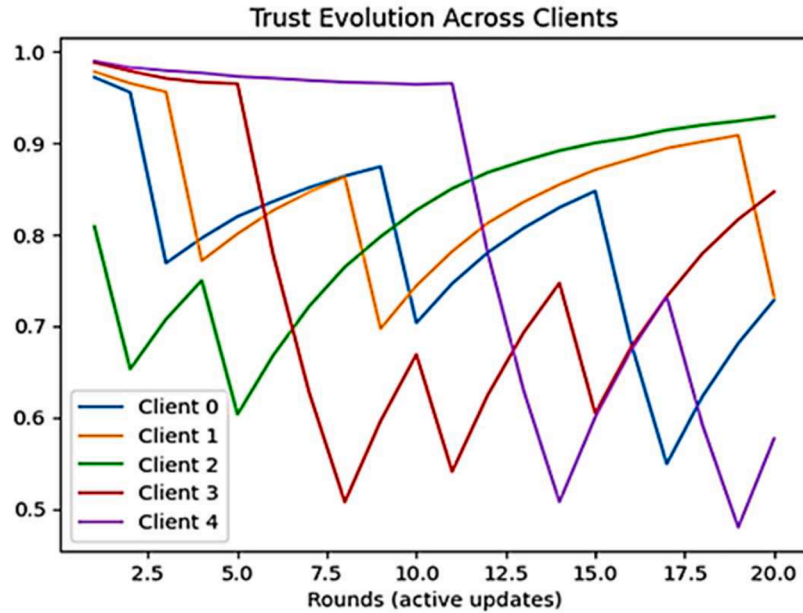


Fig. 5. Trust Evolution Across Clients under 20% Poisoning.

that robustness mechanisms do not introduce performance penalties in benign environments. At 20% poisoning, Trimmed Mean achieves the highest F1-score (0.947), followed by FedAvg (0.929) and the proposed method (0.911). This confirms that statistical filtering remains effective under moderate adversarial conditions, where malicious updates are relatively limited and easier to isolate. However, at 40% poisoning, a significant divergence emerges. FedAvg degrades sharply to 0.781, while Trimmed Mean decreases to 0.826. In contrast, the proposed trust-aware method maintains a significantly higher F1-score (0.920), corresponding to an improvement of approximately +13.9% over FedAvg and +9.4% over Trimmed Mean. These results demonstrate that the proposed approach scales more effectively under high adversarial density. This improvement can be explained by the temporal trust modeling mechanism, which accumulates behavioral evidence across rounds. Unlike statistical methods that rely on instantaneous deviation

assumptions, the proposed approach identifies persistently unreliable clients even when their updates are not extreme outliers, making it robust against stealthy and coordinated poisoning strategies.

5.1.4. Performance trade-offs under moderate attacks

Under 20% malicious client participation, a detailed evaluation of classification metrics reveals important behavioral differences between the baseline FedAvg and the proposed trust-weighted aggregation strategy, as summarized in Table 3. FedAvg achieves slightly higher overall accuracy (0.9767) and stronger precision (0.9518), indicating a more conservative decision boundary that reduces false positives. However, its recall (0.9079) is lower compared to the trust-weighted approach. In contrast, the trust-weighted method attains higher recall (0.9327), demonstrating improved sensitivity to malicious traffic and better detection of attack instances, albeit with a moderate reduction in

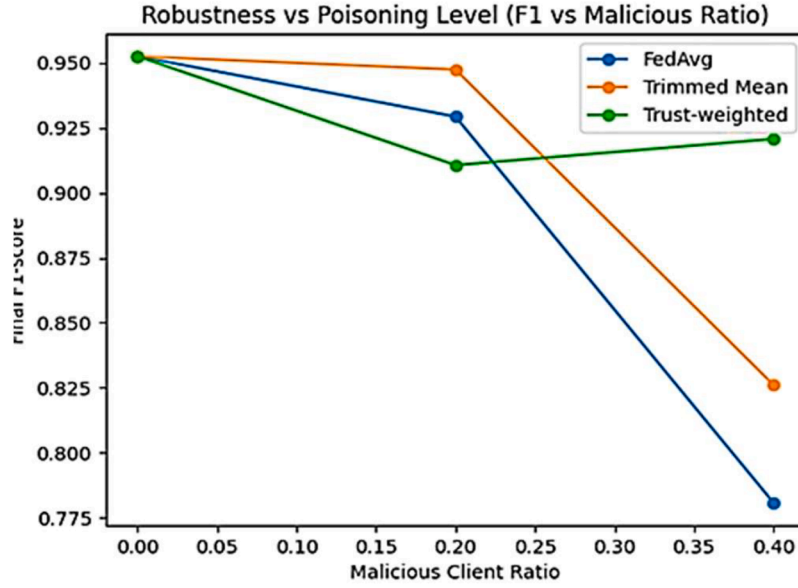


Fig. 6. F1-score vs. Malicious Ratio.

Table 2
Robustness Evaluation of Aggregation Methods under Increasing Poisoning Levels.

Malicious Ratio	FedAvg	Trimmed Mean	Trust-weighted
0%	~0.952	~0.952	~0.952
20%	0.929	0.947	0.911
40%	0.781	0.826	0.920

Table 3
Performance Metrics under 20% Poisoning.

Method	Accuracy	Precision	Recall	F1
FedAvg	0.9767	0.9518	0.9079	0.9293
Trust-weighted	0.9691	0.8895	0.9327	0.9106

precision (0.8895) and overall accuracy (0.9691). This trade-off reflects a deliberate robustness-oriented design choice: the trust-aware mechanism prioritizes attack detection sensitivity over conservative classification. In smart city security contexts—where failing to detect an attack may disrupt critical infrastructure such as transportation, energy grids, or surveillance systems—maximizing recall is often more critical than minimizing false alarms. Therefore, the observed precision-recall balance aligns with security-first deployment requirements, supporting early threat detection even at the cost of modest increases in false positives.

5.1.5. Communication efficiency

The communication cost analysis, summarized in Table 4, shows that all aggregation methods incur identical communication overhead. The proposed trust-aware mechanism does not introduce additional communication complexity, maintaining linear scaling $O(K \cdot |\theta|)$. This

Table 4
Communication Cost and Final Performance Comparison.

Method	Total Comm Cost (proxy)	Final F1
FedAvg (clean)	3034,000	0.952486
FedAvg (20% poison)	3034,000	0.929288
Trimmed Mean (20% poison)	3034,000	0.947411
Median (20% poison)	3034,000	0.922761
Trust-weighted (20% poison)	3034,000	0.910626

distinguishes it from more complex Byzantine-resilient methods that require pairwise comparisons or additional coordination, making it suitable for bandwidth-constrained smart city deployments.

5.2. Results on CSE-CIC-IDS2018

5.2.1. Convergence under domain shift

The convergence behavior on the CSE-CIC-IDS2018 dataset is shown in Figs. 7 and 8. Compared to CICIDS2017, this dataset exhibits significantly higher variability and stronger non-IID characteristics, resulting in more challenging training conditions. Under 20% poisoning, FedAvg initially reaches high accuracy (≈ 0.95 in early rounds) but subsequently exhibits strong oscillations, fluctuating between 0.89 and 0.93. These fluctuations reflect instability caused by poisoned updates interacting with heterogeneous data distributions. In contrast, Trimmed Mean and Median maintain more stable trajectories, operating within the 0.94–0.96 range. The proposed trust-aware method again demonstrates a calibration phase, maintaining moderate performance in early rounds (≈ 0.91 –0.93), followed by steady improvement after trust stabilization. As shown in Fig. 7, the method converges to approximately 0.947 accuracy by Round 20 with reduced variance, indicating improved robustness under heterogeneous conditions.

5.2.2. F1-score dynamics and robustness

The F1-score evolution in Fig. 8 provides a more informative perspective due to class imbalance. FedAvg exhibits strong instability, with values dropping to ≈ 0.63 in intermediate rounds before partially recovering to ≈ 0.82 . This irregular behavior highlights its vulnerability to adversarial updates. Trimmed Mean and Median show more stable performance, maintaining F1-scores in the range of 0.80–0.89. The proposed trust-aware method initially lags (≈ 0.73 –0.76) due to trust calibration but steadily improves to ≈ 0.84 –0.86 with significantly reduced oscillations. This confirms that dynamic trust recalibration effectively suppresses adversarial influence over time.

5.2.3. Robustness under high adversarial density

Under 40% poisoning, the performance comparison is summarized in Table 5 and illustrated in Figs. 9 and 10. FedAvg drops to an F1-score of 0.7826, while Trimmed Mean degrades further to 0.7512. Median maintains moderate robustness (0.8143), but still exhibits performance loss. In contrast, the proposed trust-aware method achieves the highest F1-score (0.8514) and accuracy (0.9480), corresponding to an

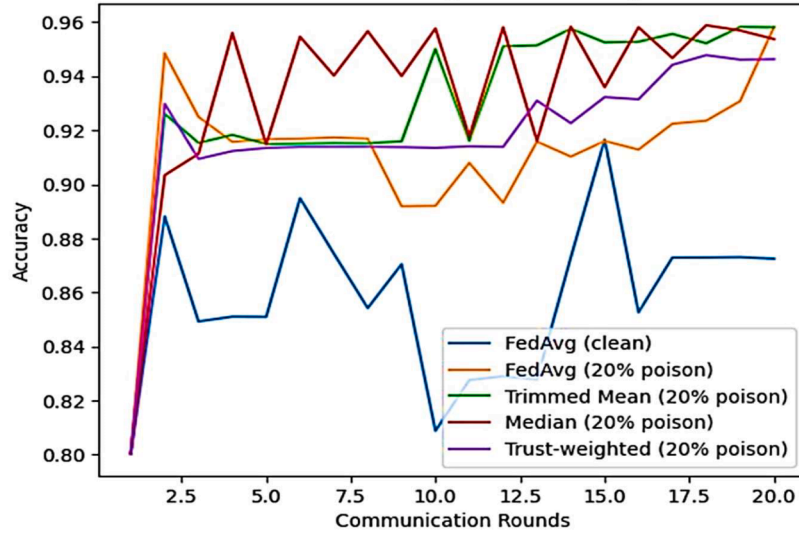


Fig. 7. Accuracy vs. Communication Rounds.

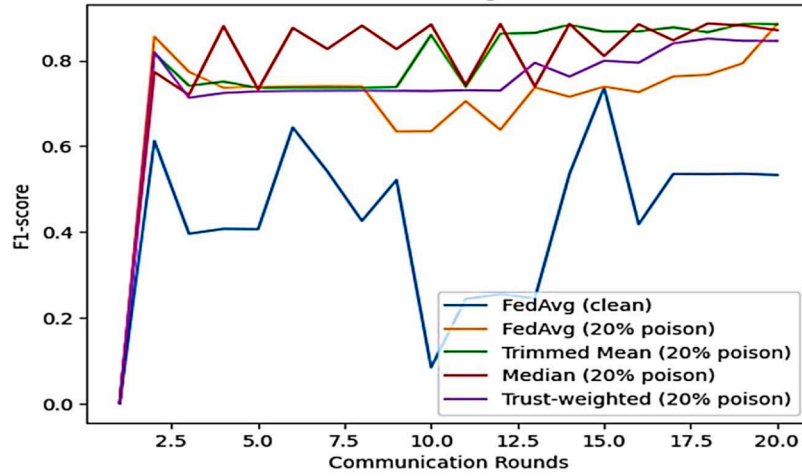


Fig. 8. F1-score vs. Communication Rounds.

Table 5

Final Results @ Round 20 (40% Poisoning).

Method	Final Accuracy	Final F1
FedAvg	0.9216	0.7826
Trimmed Mean	0.8891	0.7512
Median	0.9236	0.8143
Trust-weighted	0.9480	0.8514

improvement of approximately +8.8% over FedAvg and +13.3% over Trimmed Mean. These results confirm that the advantage of trust-aware aggregation becomes more pronounced as adversarial density increases. The convergence curves in Figs. 9 and 10 further reinforce this observation. While baseline methods exhibit instability and degradation, the trust-aware method stabilizes after initial calibration and maintains a consistent upward trajectory, demonstrating superior resilience.

5.2.4. Trust-based client differentiation

The final trust scores in Table 6 provide additional insight into the mechanism's effectiveness. Malicious clients are assigned near-zero trust values (e.g., 0.0115 and 0.0546), while benign clients maintain high trust (up to 0.9986). This clear separation confirms that the framework successfully distinguishes reliable from unreliable

participants based solely on behavioral consistency.

5.2.5. Cross-dataset analysis and key insights

The comparative analysis across CICIDS2017 and CSE-CIC-IDS2018 reveals important insights into the behavior of federated aggregation strategies under varying data conditions. First, the results demonstrate that statistical aggregation methods perform well under moderate adversarial conditions but degrade significantly as adversarial density and data heterogeneity increase. This limitation stems from their reliance on instantaneous deviation assumptions, which become unreliable when benign updates naturally diverge in non-IID environments. Second, the proposed trust-aware mechanism maintains stable performance across both datasets, indicating strong generalization capability. Unlike statistical methods, it decouples aggregation weights from purely geometric assumptions and instead relies on temporal behavioral consistency, which remains robust under domain shifts. Third, the results highlight a key trade-off between convergence speed and robustness. While FedAvg and statistical methods achieve faster initial convergence, they are more susceptible to instability under adversarial conditions. In contrast, the trust-aware method introduces a short calibration phase but achieves superior long-term stability and robustness. Overall, the experimental findings demonstrate that trust-aware aggregation provides a scalable and adaptive solution for federated intrusion detection,

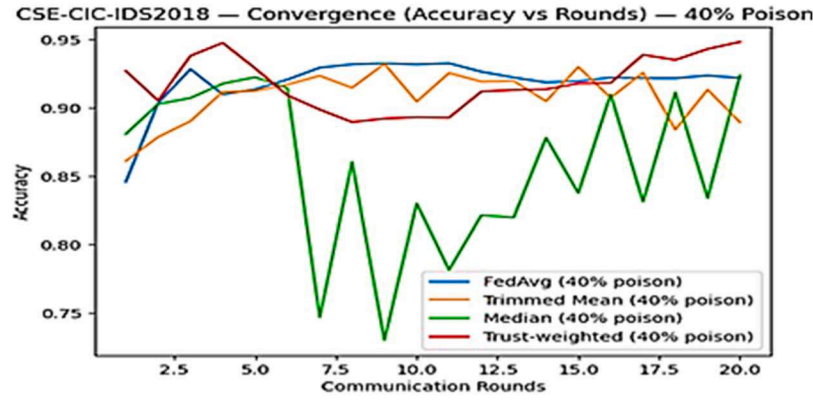


Fig. 9. Accuracy vs. Rounds — 40% Poisoning.

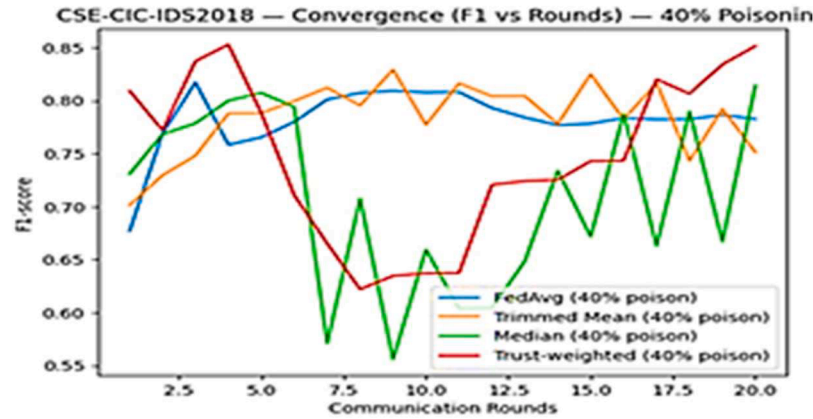


Fig. 10. F1 vs. Rounds — 40% Poisoning.

Table 6

Final Trust Scores (Round 20 — 40% Poisoning).

Client ID	Final Trust Score	Reliability Interpretation
Client 1	0.6603	High (Benign)
Client 2	0.0546	Very Low (Malicious)
Client 3	0.9986	Extremely High (Benign)
Client 4	0.4873	Moderate / Partially Reliable
Client 5	0.0115	Near-Zero (Malicious)

particularly in realistic smart city environments characterized by heterogeneous data distributions and high adversarial risk.

6. Discussion and comparison with existing work

The proposed framework differs from existing trust-aware federated learning approaches in three major aspects: (1) it does not require a trusted server-side reference dataset as assumed in FLTrust, (2) it models client reliability through temporal behavioral consistency rather than

Table 7

Comparison with Existing Federated Learning-Based Intrusion Detection.

Work (Authors, Year)	Dataset(s)	FL-Based IDS	Robust to Poisoning	Non-IID Handling	Trust Mechanism	Key Results	Limitation
McMahan et al., 2017 [46]	MNIST, CIFAR	Yes	No	Limited	No	High accuracy (clean)	No adversarial robustness
Blanchard et al., 2017 (Krum) [50]	Synthetic / ML	Yes	Yes	Assumes bounded divergence	No	Byzantine resilient	Fails under strong non-IID
Yin et al., 2018 (Median/Trimmed Mean) [51]	ML benchmarks	Yes	Yes	May over-filter	No	Stable under attacks	Degrades at high poisoning
Fung et al., 2018 (FoolsGold) [52]	Synthetic	Yes	Partial	Limited	Yes	Sybil defense	Not general poisoning defense
Cao et al., 2020 (FLTrust) [53]	MNIST, CIFAR	Yes	Yes	Depends on trusted data	Yes	Strong robustness	Requires trusted dataset
Kairouz et al., 2021 [54]	Multiple	Yes	No	Discussed	No	Comprehensive FL survey	No IDS robustness
ALBANBAY et al., 2025 [47]	CICIDS2017, IoT	Yes	No	Partial	No	Improved detection	No poisoning robustness
Akter et al., 2026 [48]	UNSW-NB15	Yes	No	Partial	No	F1 ~0.97	Focus on privacy
This Work (Houichi et al.)	CICIDS2017, CSE-CIC-IDS2018	Yes	Yes (20–40%)	Explicit	Dynamic	F1 ~0.92 / 0.85	Calibration phase

only instantaneous gradient similarity as in FoolsGold, and (3) it explicitly targets smart city intrusion detection environments characterized by strong non-IID traffic heterogeneity and persistent adversarial participation. This combination enables improved robustness under realistic smart city deployment conditions while preserving communication efficiency and adaptability. To position the proposed framework within the broader landscape of robust federated learning and intrusion detection, Table 7 presents a comparative analysis of representative state-of-the-art approaches, including classical federated learning methods, Byzantine-resilient aggregation techniques, trust-based mechanisms, and recent federated intrusion detection systems [46]. Early federated learning frameworks such as FedAvg demonstrate strong performance under benign conditions; however, they lack robustness against adversarial clients and exhibit limited capability in handling non-independent and identically distributed (non-IID) data [50]. This limitation becomes particularly critical in smart city environments, where data heterogeneity and adversarial behavior are inherent characteristics. To address adversarial robustness, Byzantine-resilient aggregation methods such as Krum and coordinate-wise Median or Trimmed Mean have been proposed [51]. These approaches improve resilience by filtering abnormal updates based on statistical assumptions. Nevertheless, their effectiveness depends on restrictive conditions such as bounded divergence or symmetric contamination, which are often violated in realistic IoT deployments. As a result, their performance tends to degrade under strong non-IID conditions or high malicious participation. Trust-based federated learning approaches introduce adaptive mechanisms to evaluate client reliability. For instance, FoolsGold [52] mitigates Sybil attacks by penalizing highly similar updates, but relies on update similarity assumptions that may not hold in heterogeneous network traffic scenarios. Similarly, FLTrust [53] achieves strong robustness by leveraging a trusted server-side dataset; however, such a requirement limits its applicability in decentralized and privacy-sensitive smart city environments. In addition, comprehensive studies such as Kairouz et al. [54] provide an overview of federated learning challenges, including statistical heterogeneity, but do not offer concrete solutions tailored to intrusion detection robustness. Recent federated learning-based intrusion detection systems, including Albanbay et al. [47] and Akter et al. [48], demonstrate improved detection performance and privacy preservation on datasets such as CICIDS2017 and UNSW-NB15. Recent advances in robust federated learning have increasingly explored adaptive trust management, blockchain-assisted reputation systems, and secure collaborative learning to improve resilience against sophisticated adversarial attacks in distributed environments. While blockchain-based approaches provide decentralized trust management and immutable reputation records, they often introduce additional communication, storage, and computational overhead that may not be suitable for latency-sensitive and resource-constrained smart city IoT deployments. Similarly, many adaptive trust mechanisms rely on trusted server-side reference datasets, external validation resources, or complex reputation infrastructures that may be impractical in privacy-preserving federated learning scenarios. In contrast, the proposed framework estimates client reliability directly from behavioral consistency during the federated learning process, requiring neither trusted auxiliary datasets nor blockchain support. This lightweight design preserves communication efficiency while maintaining robustness against malicious model updates under heterogeneous non-IID environments, making it well suited for large-scale operational smart city deployments. However, these approaches primarily focus on detection accuracy and do not explicitly address adversarial robustness or poisoning attacks. Consequently, their applicability in hostile environments remains limited. In contrast, as summarized in Table 7, the proposed trust-aware framework simultaneously addresses non-IID data heterogeneity and adversarial behavior through dynamic trust-based aggregation. Unlike prior methods that rely on static filtering rules or strong assumptions, the proposed approach adaptively adjusts client influence based on longitudinal behavioral consistency, without

requiring trusted validation data. Furthermore, experimental results on CICIDS2017 and CSE-CIC-IDS2018 demonstrate that the proposed method maintains stable performance under both moderate and severe poisoning scenarios. In particular, under high adversarial participation (e.g., 40% malicious clients), the framework exhibits improved robustness compared to classical aggregation strategies, while preserving convergence stability and communication efficiency. This combination of adaptive robustness, realistic deployment modeling, and cross-dataset validation distinguishes the proposed approach as a practical and generalizable solution for secure federated intrusion detection in smart city environments.

The proposed framework contributes to secure federated intrusion detection by reducing the influence of malicious or unreliable participants through dynamic trust-aware aggregation. Unlike conventional Federated Averaging, which remains highly vulnerable to poisoned model updates, the proposed approach introduces temporal trust evolution and deviation-aware filtering to progressively suppress adversarial influence across communication rounds. This design aligns with prior research on poisoning attacks and adversarial machine learning defenses, which emphasizes the importance of robust aggregation and adaptive reliability estimation in distributed learning systems [55]. In addition, recent advances in lightweight cryptographic implementations and secure digital signature architectures, including optimized Ed25519/Ed448 hardware implementations and FPGA-based post-quantum cryptographic accelerators, highlight the growing importance of combining federated learning robustness with efficient cryptographic protection mechanisms for resource-constrained smart city infrastructures [56–58]. From a theoretical perspective, the proposed trust-aware aggregation mechanism is expected to improve convergence stability by progressively adapting the influence of participating clients according to their long-term behavioral consistency. The exponential trust update introduces temporal smoothing that limits abrupt variations in aggregation weights, thereby reducing instability caused by transient statistical heterogeneity or noisy local updates. As communication rounds progress, persistently reliable clients receive increasingly larger aggregation weights, whereas malicious or inconsistent clients are gradually down-weighted, leading to a more stable optimization trajectory than conventional equal-weight aggregation. Although a rigorous mathematical convergence proof under heterogeneous non-IID data and adaptive adversarial settings remains beyond the scope of this work, the observed convergence behavior across both benchmark datasets is consistent with this theoretical intuition. Establishing formal convergence guarantees and robustness bounds for adaptive trust-aware federated aggregation represents an important direction for future research. From a practical deployment perspective, the proposed framework can be integrated into existing smart city cybersecurity infrastructures following a hierarchical edge–cloud architecture. IoT devices deployed across transportation systems, smart grids, healthcare facilities, surveillance networks, and environmental monitoring platforms continuously generate network traffic that is processed locally by nearby edge gateways. These gateways perform feature extraction, local intrusion detection model training, and trust-aware participation in the federated learning process without transmitting raw traffic data, thereby preserving data privacy and reducing communication costs. A centralized federated server coordinates global model aggregation and trust management, while the Smart City Security Operations Center (SOC) supervises model performance, security alerts, and policy updates. In practical deployments, challenges such as heterogeneous computing capabilities, intermittent client participation, communication latency, bandwidth limitations, and dynamic network conditions must be considered. Since the proposed framework modifies only the aggregation strategy without altering the standard federated learning communication protocol, it can be incorporated into existing federated learning infrastructures with minimal deployment complexity while preserving communication efficiency and scalability. Despite the promising results, several limitations remain. First, the proposed trust estimation

mechanism requires an initial calibration phase during early communication rounds, which may temporarily slow convergence compared with simpler aggregation approaches. Second, the effectiveness of the framework depends on hyperparameter selection, including the trust smoothing factor, deviation thresholds, and clipping bounds. Third, the current experimental evaluation mainly focuses on poisoning attacks under non-IID conditions and does not yet comprehensively evaluate advanced adversarial scenarios such as backdoor attacks, adaptive stealthy poisoning, or large-scale Sybil coordination strategies. Furthermore, although the framework maintains communication efficiency comparable to standard federated learning, additional computational overhead may appear in large-scale deployments involving complex trust monitoring and real-time edge orchestration. Although the experimental evaluation presented in this work focuses on Byzantine poisoning attacks, the proposed trust-aware aggregation mechanism is not inherently limited to this adversarial model. Because client trust is updated continuously according to long-term behavioral consistency rather than a single communication round, the framework is expected to progressively reduce the influence of participants exhibiting persistent malicious behavior, including stealthy poisoning and coordinated attack strategies. Similarly, clients participating in Sybil attacks are expected to experience decreasing trust when their cumulative behavior consistently deviates from reliable participants. Nevertheless, comprehensive quantitative evaluation under backdoor attacks, adaptive stealthy poisoning, and coordinated Sybil attacks is required to fully characterize the robustness of the proposed framework. These investigations constitute an important direction for future work and will further strengthen the practical applicability of the proposed approach in real-world smart city IoT environments. Future research directions include extending the proposed framework to evaluate robustness against advanced adversarial scenarios such as backdoor attacks, stealthy poisoning, and coordinated Sybil adversaries, while also integrating secure aggregation, lightweight cryptographic authentication, and post-quantum security mechanisms to achieve end-to-end trustworthy federated intrusion detection. In addition, future work will investigate theoretical convergence guarantees, adaptive trust calibration strategies, and practical deployment challenges in real-world smart city environments, including latency, energy consumption, scalability, and long-term trust evolution. Another important consideration is the selection of the framework hyperparameters, including the trust smoothing factor (λ), the weighting parameter (α), and the outlier detection threshold. These parameters govern the balance between trust stability, responsiveness to behavioral changes, and the sensitivity of malicious client detection. In this study, fixed parameter values were adopted throughout all experiments to ensure consistent and fair comparisons across the evaluated aggregation methods. Empirically, the proposed framework demonstrated stable performance under the selected configuration across both benchmark datasets and different adversarial participation levels. Nevertheless, the influence of these parameters may vary depending on the degree of statistical heterogeneity, attack intensity, and client participation patterns. Therefore, a comprehensive quantitative sensitivity and ablation analysis covering a wider range of parameter settings constitutes an important direction for future work and may provide additional guidance for adapting the framework to different operational smart city environments.

7. Conclusion and future work

This paper presented a trust-aware federated intrusion detection framework for smart city IoT environments, addressing the dual challenges of data heterogeneity (non-IID distributions) and adversarial participation in collaborative learning systems. The proposed approach introduces a dynamic trust evaluation mechanism that continuously assesses client reliability based on validation consistency and update deviation, and integrates this information into the aggregation process through trust-weighted contributions, outlier filtering, and optional

update clipping. The experimental results demonstrate that the proposed framework achieves a robust and adaptive behavior across different operating conditions. Under moderate adversarial scenarios, the method maintains competitive performance compared to statistical robust aggregation techniques. More importantly, under high adversarial participation (e.g., 40% malicious clients), the trust-aware aggregation mechanism significantly improves robustness and stability, effectively mitigating the impact of malicious updates while preserving convergence. This improvement is attributed to the temporal modeling of client reliability, which enables the system to identify persistently unreliable participants even when their updates are not extreme outliers. Furthermore, the framework exhibits consistent performance across multiple datasets, confirming its ability to generalize under varying traffic distributions and levels of heterogeneity. In addition, the proposed solution does not introduce additional communication overhead compared to standard federated learning, making it suitable for large-scale smart city deployments with constrained network resources. Despite these advantages, several limitations remain. The trust estimation process requires an initial calibration phase during early communication rounds, which may slightly delay convergence compared to simpler aggregation methods. In addition, the effectiveness of the framework depends on the selection of hyperparameters, such as the trust smoothing factor and deviation thresholds, which may require adaptation to different deployment contexts. Future work will focus on extending the proposed framework along several directions. First, more advanced adversarial models, including backdoor attacks, stealthy poisoning, and coordinated Sybil strategies, will be investigated to further evaluate robustness. Second, theoretical analysis of convergence and robustness guarantees under dynamic trust weighting will be explored. Third, the integration of privacy-enhancing techniques such as secure aggregation and differential privacy will be studied to jointly address confidentiality and robustness requirements. Finally, real-world deployment and large-scale validation in operational smart city environments will be pursued to assess practical constraints, including latency, communication efficiency, and long-term trust evolution.

CRedit authorship contribution statement

Mehdi Houichi: Supervision, Software, Project administration, Methodology, Formal analysis, Conceptualization. **Faouzi Jaidi:** Supervision, Methodology, Validation, Writing – review & editing, Resources. **Adel Bouhoula:** Supervision, Conceptualization, Methodology.

Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. This research was conducted independently and without any commercial or financial support that could be construed as a potential conflict of interest.

Data availability

Open source data

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